Ollscoil Mhá Nuad
Maynooth University

SEMESTER 1
2025-2026

CS320
Computer Networks

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Time allowed: 2 hours

You must answer Question 1 and any three other questions.
Your mark will be based on Question 1 and your best three answers from the remaining questions

All questions carry equal marks

Instructions

	Yes	No	N/A
Formulae and Tables book allowed *(i.e. available on request)*		X	
Formulae and Tables book required *(i.e. distributed prior to exam commencing)*		X	
Statistics Tables and Formulae allowed *(i.e. available on request)*		X	
Statistics Tables and Formulae required *(i.e. distributed prior to exam commencing)*		X	
Dictionary allowed *(supplied by the student)*		X	
Non-programmable calculator allowed	X		
Students required to write in and return the exam question paper		X	

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[25 marks]

1 Short Questions

In a few lines please answer the following short questions (no essays!!)

(a) A 2 MB file is sent between two hosts separated by 4 routers. Each link has a rate of 10 Mbps, a propagation delay of 5 ms, and uses store-and-forward switching. Ignoring queueing and processing, calculate the total end-to-end delay. [2.5 marks]

(b) In what situation would you prefer packet switching over circuit switching? [2.5 marks]

(c) What is the purpose of the DNS system, and where are its records stored? [2.5 marks]

(d) What does it mean for a protocol such as HTTP to be stateless? Give one way in which HTTP can maintain state when needed. [2.5 marks]

(e) List four services that TCP provides to applications that UDP does not and one they both provide. Why would anyone ever bother with UDP? [2.5 marks]

(f) Explain the difference between routing and forwarding in a router. [2.5 marks]

(g) Suppose a router has a forwarding entry 192.168.0.0/24 → Interface 1. Which destination IP addresses will match this entry? [2.5 marks]

(h) What is the role of the Address Resolution Protocol (ARP)? [2.5 marks]

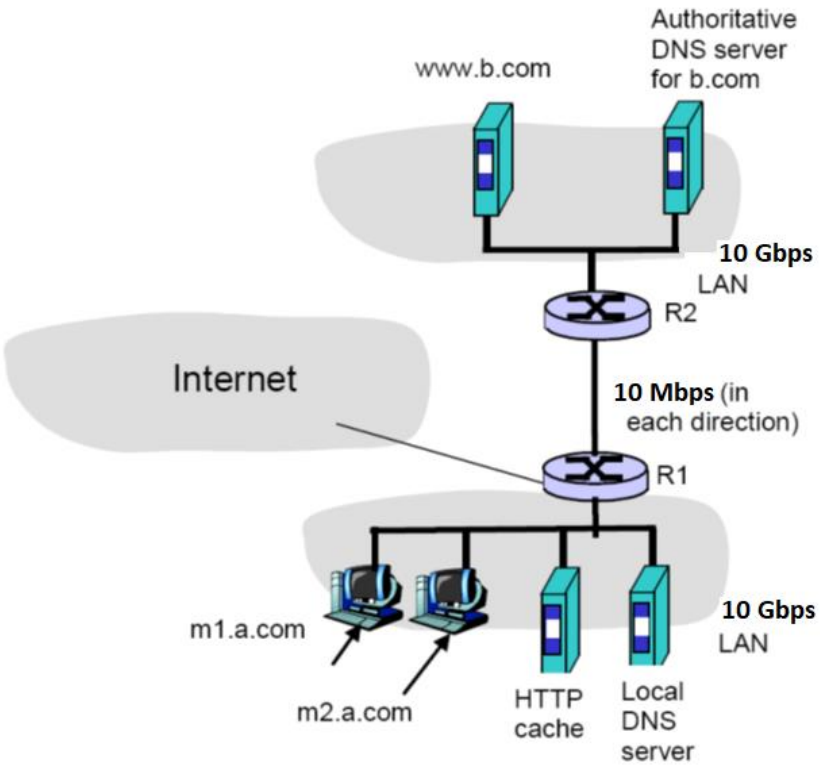
(i) What is the difference between CSMA and token passing as medium-access methods? [2.5 marks]

(j) What is the purpose of a digital signature? Briefly explain how it provides authentication. [2.5 marks]

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2 **Delays, Throughput and Caches**

Consider the networks shown in the figure. There are two user machines **m1.a.com** and **m2.a.com** in the network **a.com**. Suppose the user at **m1.a.com** types in the URL **www.b.com/bigfile.htm** into a browser to retrieve a 2Gbit (2000 Mbit) file from **www.b.com**.



(a) List the sequence of DNS and HTTP messages sent/received from/by m1.a.com as well as any other messages that leave/enter the a.com network that are not directly sent/received by m1.a.com from the point that the URL is entered into the browser until the file is completely received. Indicate the source and destination of each message. You can assume that every HTTP request by m1.a.com is first directed to the HTTP cache in a.com and that the cache is initially empty, and that all DNS requests are iterated queries – assume DNS cache is also empty at the start. *Note, if a DNS query needs to be resolved it can be done so via iterated DNS to root, TLD, and authoritative DNS server.* [7 marks]

(b) How much time does it take to accomplish the steps you outlined in your answer to (a)? Explain how you arrived at this answer. In answering this question, you can make the following assumptions [7 marks]

- The packets containing any DNS commands and HTTP commands such as GET are very small compared to the size of the file, and thus their transmission times (but not their propagation times) can be neglected.

- Propagation delays within the LAN are small enough to be ignored. The propagation from router R1 to router R2 is 100 ms.
- The propagation delay to/from anywhere in a.com to any other site on the Internet (except b.com) is 500 ms.
- You can neglect TCP hand-shaking delays for the HTTP exchanges. Keep things simple!

(c) Now assume that machine m2.a.com makes a request to exactly the same URL that m1.a.com made. List the sequence of DNS and HTTP messages sent/received from/by m2.a.com as well as any other messages that leave/enter the a.com network that are not directly sent/received by m2.a.com from the point that the URL is entered into the browser until the file is completely received. [6 marks]

(d) How much time does it take to accomplish the steps that you outlined in your answer to (c)? [5 marks]

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[25 marks]

3 TCP

Answer questions a) – d) below briefly, in just a few sentences. Question e) will take some more time and thought and require a longer answer.

(a) What is the purpose of the receiver-advertised window rwnd in TCP? [5 marks]

(b) Identify 6 fields in the TCP header, and give a one-sentence description of each of their uses. [5 marks]

(c) Consider two TCP sessions that must share a link's bandwidth. One of the TCP connections has been running for quite some time and has built up a large TCP sending window. The second connection then starts up with an initially small window. Long term, what will be the relative throughput achieved by these two TCP sessions? Explain. [5 marks]

(d) Consider a TCP session and a UDP session that must share a link's bandwidth. Of course, both sessions would ideally like to send as fast as they can. Long term, what will be the relative throughput achieved by these two sessions. Explain.. [5 marks]

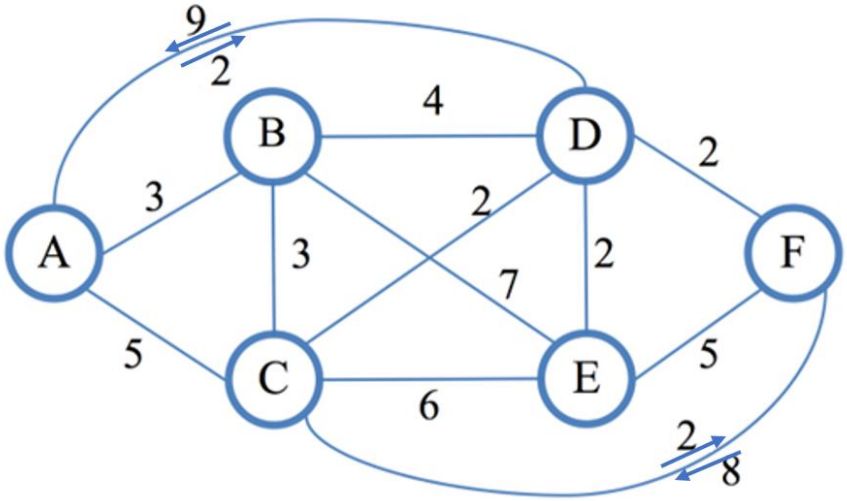
(e) A TCP sender transmits a segment with sequence number 1000 and 1000 bytes of data. The receiver acknowledges it with ACK 2000. The sender then sends two more segments: one from 2000–2999 and another from 3000–3999. The second of these segments (3000–3999) is lost. Explain what happens next, describing the acknowledgements sent by the receiver and how the sender uses duplicate ACKs to detect and recover from the loss. Would TCP Taho and Reno behavie differently in this scenario? [5 marks]

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[25 marks]

4 Routing Algorithms

Consider the network with nodes A–F shown in the figure below. Link costs are shown on each directed link. Note that two of the links (A–D and C–F) are asymmetric, i.e. the cost from A to D is different from the cost from D to A, and similarly for C and F.



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(a) Using Dijkstra's algorithm, with A as the source, compute the shortest-path tree. Show your working step-by-step in a table, listing after each iteration: the set of nodes N′ for which the least-cost path from A is known; and for each node X ∈ {B, C, D, E, F}, the pair (c(X), p(X)), where c(X) is the current known cost from A to X and p(X) is the predecessor of X on that path.. [17 marks]

(b) Using your result from part (a), what is the shortest path from A to F (node by node)? Briefly explain how you obtained your answer from the Dijkstra table. [3 marks]

(c) Briefly explain the primary differences between link-state and distance-vector routing algorithms. [5 marks]

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[25 marks]

5 Link Layer and Network Layer Addressing

(a) An ISP owns the address block 200.25.64.0/18. The ISP must allocate address space to five customers A–E, who require the following numbers of usable IP addresses: [10 marks]

Customer	Required Addresses
A	3000
B	6000
C	2000
D	9000
E	4000

Address ranges are allocated first to A, then B, then C, then D (in that order), using the lowest available starting address each time. Any remaining unallocated block is given to E. Determine the address range and subnet mask allocated to each organization, using CIDR notation (e.g. 200.25.64.0/18).

(b) A host X with IP address 200.25.80.10 in Customer B's network sends an IP packet to host Y at 200.25.70.25 in Customer A's network. Both networks are connected to their own ISP router on their own Ethernet LANs. State the Ethernet source and destination addresses and the IP source and destination addresses on: [6 marks]

(i) the link between host X and the ISP router, and
(ii) the link between the ISP router and host Y.

(c) Host P sends an ARP query to learn the MAC address of host Q on the same subnet. Assume all switches are learning switches with empty forwarding tables when the query is sent. How many nodes receive the ARP request, and how many nodes receive the ARP reply? Explain briefly. [5 marks]

(d) A company implements **port-based VLANs** on a single Ethernet switch. Explain briefly what a port-based VLAN is, how it provides traffic isolation, and how communication between VLANs can occur. [4 marks]

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[25 marks]

6 Security: Authentication Protocols

(a) Consider the authentication protocol in the figure below, in which Alice authenticates herself to Bob using shared secret key K_{A-B} as shown. In some systems, both parties need to verify each other's identity — this is known as *mutual authentication*. In such a case, Bob would also run the same protocol in reverse to authenticate himself to Alice, possibly at the same time. Explain why this mutual-authentication version of the protocol is insecure, and describe briefly how an attacker Trudy could exploit the weakness to convince Bob that she is Alice. (Hint: pay close attention to how each side uses its nonce and how the same nonce might be reflected or reused maliciously.) [20 marks]

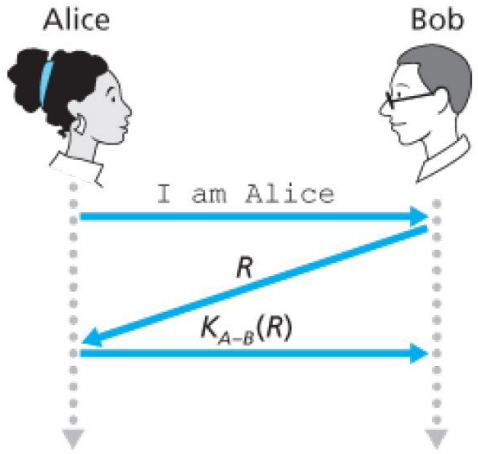


Figure X Simple Authentication Protocol

(b) Suppose Alice and Bob are communicating over an SSL session. Suppose an attacker, who does not have any of the shared keys, inserts a bogus TCP segment into a packet stream with correct TCP checksum and sequence numbers (and correct IP addresses and port numbers). Will SSL at the receiving side accept the bogus packet and pass the payload to the receiving application? Why or why not? [5 marks]